Kurayi Chawatama

Bursa/Türkiye kurichawaz@gmail.com +90 505 650 69 12 Website

EDUCATION

Molecular Biology and Genetics, B.Sc.*(In Turkish)* | Uludağ University, Türkiye | Oct 2023 – July 2027 | GPA: 3.27/4 *(Fully Funded Turkish Government Scholarship)*

RESEARCH & PROFESSIONAL EXPERIENCE

Bioinformatics Pipeline Developer Oct 2026 – Present

Genalyse UK Ltd | United Kingdom (Remote)

- Developing scalable Nextflow pipelines for clinical RNA-seq analysis, integrating QC, STAR alignment, Salmon quantification, differential expression, GSEA, aberrant splicing, expression outlier detection, and allele-specific expression.
- Engineering backend-ready integration through structured data contracts, AWS compute/cost optimization, and UI/UX mockups to guide frontend development.

Final Year Research Project Oct 2026 – Jan 2027

Assoc. Prof. Dilek Pirim | Uludağ University, Türkiye

- Selected as 1 of 8 students from a class of 65 to propose and undertake a semester-long research project culminating in a thesis report and presentation.
- Benchmarking statistical methods (Spearman correlation, Linear regression, MaAsLin2, MaAsLin3, MITScan) for detecting host–microbiome associations using metagenomic and host RNA-seq data from Inflammatory Bowel Disease (IBDMDB), evaluating statistical performance, robustness, and computational efficiency.

Undergraduate Researcher Oct 2024 – Present

Prof. Berrin Tunca | Uludağ University, Türkiye

- *"Characterizing Tumor Microenvironment Transcriptome Variations in NASH and Viral Hepatocellular Carcinomas for Targeted Therapy."*
- Proposed and secured a grant to perform cross-species scRNA-seq integration and differential analysis of the poorly characterized and therapeutically challenging NASH-HCC, which lacks human scRNA-seq data, and validated in vitro using an FFA-induced HCC model. *(TÜBİTAK 2209-A Grant Funded)*
- Identified potential therapy resistance mechanisms involving hepatocyte metabolic rewiring, stromal activation, and immune–stromal crosstalk in MASH-HCC.
- The preliminary results poster was awarded the honor prize at a bioinformatics symposium and was later presented at an international immunology symposium.

Undergraduate Researcher Jul 2026 – Sep 2026

Assoc. Prof. Ogün Adebali | Sabancı University, Türkiye

- Developed a phylogenomics framework to map mutational tolerance and nucleotide-level evolution rates across pathogenic bacterial genomes.
- This was used to identify fast-evolving regions that correlate with antimicrobial drug resistance and slow-evolving regions that contain potential therapeutic targets.
- Built a scalable Snakemake pipeline for MSA, phylogeny reconstruction, and evolution rate scoring for pathogens. *(Türkiye Scholarships Funded Internship)*

Bioinformatics Intern Jan 2026 – Feb 2026

Asst. Prof. Burçak Otlu | Middle East Technical University, Türkiye (Remote)

- Analyzed large-scale cancer genome datasets to identify mutations and mutational processes associated with tumor development.
- Worked with Nextflow pipelines for large-scale WES/WGS cancer genomics, including somatic variant calling and mutational signature analysis.
- Performed basic processing and analysis of multiple omics data types at scale, including: ChIP-seq, Methyl-seq, and Bulk RNA-seq. *(EU-funded SUSA Internship)*

Undergraduate Researcher Jul 2025 – Oct 2025

Prof. Vicente Pelechano Garcia | Karolinska Institutet, Sweden

- Developed a regression-based method to score cancer cell plasticity and decode its gene signature in Hepatocellular Carcinoma (Huh7) using CITE-seq data.
- Validated the cell plasticity signature in the TCGA-LIHC clinical cohort by Cox regression analysis, linking it to drug resistance, relapse, and survival.
- Identified distinct non-genetic phenotypic states in clonal Huh7 cell populations using scRNA-seq. *(Erasmus+ Funded Traineeship)*

Research Intern Jul 2024 – Sep 2024

Asst. Prof. Atakan Ekiz | Izmir Institute Of Technology, Türkiye

- Performed bioinformatics analyses in tumor immunology using bulk/single-cell RNA-seq, TCR-seq, and microarray data to investigate immune-related lncRNAs.
- Assisted with lentiviral transduction, cell culture passaging, mouse xenograft modeling, and flow cytometry.

Undergraduate Researcher & Intern May 2023 – Jul 2024

Asst. Prof. Elif Uz Yıldırım | Uludağ University, Türkiye

- *"Comprehensive Analysis of GPRC5A Promoter Methylation Across Cancer Types."* Methyl-seq pilot study poster presented at a bioinformatics symposium.
- Carried out routine molecular genetics techniques: DNA isolation, PCR, and Gel Electrophoresis.

VOLUNTEERING & COMMUNITY INVOLVEMENT

Contributor & Core Team Member Mar 2025 – Present

nf-core

- nf-core is an open-source bioinformatics initiative dedicated to building peer-reviewed, reproducible data analysis pipelines used by millions of researchers globally.
- Ranked top 3 code contributor to the Single Cell RNA-seq analysis pipeline, resolving runtime errors and engineering a comprehensive doublet-detection module.
- Authored documentation and a blog post on running nf-core pipelines in Google Colab, leading to my induction into the core outreach team (managing committee).

Task Force Member Oct 2024 – Present

Young European Federation of Immunological Societies (yEFIS)

- Organized a webinar exploring immunoinformatics applications across industry and academia with the Immune Collaborators workgroup.
- Conceptualized and led development of an open-access blog page to be run by the Immune Collaborators workgroup for the community to democratize technical knowledge, managing an editorial pipeline of experimental advice and career-navigation resources for early-career researchers in immunology.

Outreach & Translation Team Leader Jan 2024 – Present

International Society for Computational Biology - RSG Türkiye

- Organized a hybrid (online and in-person) scRNA-seq Hackathon and a careers Q&A panel with RSG Australia.
- Coordinated the Türkiye local hubs of the international 2025 and 2026 nf-core hackathon, securing a venue and designing projects for participants of all skill levels.
- Directed Turkish translation and hosting on our webpage of bioinformatics learning resources to increase accessibility for students in Türkiye.
- Developing a free, 5-week bioinformatics (RNA-seq-based) course to expand access to hands-on training in Turkish; resources are currently scarce and paywalled.

Science Team Leader Nov 2022 – Present

UluRover | Uludağ University Robotics Society

- Led R&D for an interplanetary rover competing on simulated Martian surfaces. Integrated chemical & robotics modules into an autonomous life detection minilab.
- Helped pitch the project to multiple industry partners and secure over $10,000 in materials and sponsorships.
- Received key career mentorship from the previous team leader and subsequently institutionalized annual recruitment and mentoring of two life sciences freshmen.
- Led the science team in an additional project that won Best Presentation & Pitch at the 2025 Teknofest Biotechnology Startup Idea Competition. Placed 5th/1400.

Founding Board Member Mar 2025 – Present

Zimbabwe Students in Türkiye Association (ZSTA)

- Co-founded with 6 other students in coordination with the Embassy of Zimbabwe, establishing a strategic network that has quickly grown to serve ~100 students.
- Provided academic mentorship, coordinated a career guidance seminar, an onboarding session for new students, and the 2026 Independence Day celebration.
- Cultivated institutional relationships with Turkish industry partners to guarantee internship spots for members studying engineering disciplines.

AWARDS & SCHOLARSHIPS

- Türkiye Scholarship from the Turkish Government for fully funded undergraduate studies (2022).
- Scientific and Technological Research Council of Türkiye (TÜBİTAK) 2209-A undergraduate research project grant (2025).
- European Union-funded Erasmus+ traineeship grant for research at Karolinska Institutet, home of the Nobel Prize in Physiology & Medicine (2025).
- European Union-funded Sustainable Healthcare with Digital Health Data Competence (SUSA) winter internship at Middle East Technical University (2026).
- Türkiye Scholarships-funded PURE summer research internship at Sabancı University, 240 selected from 10 173 international applicants (2026).
- Poster Presentation Honor Prize awarded at the 18th International Symposium on Health Informatics and Bioinformatics (2025).
- Best Presentation & Pitch Award awarded at the TEKNOFEST Biotechnology Startup Idea Competition among 1400 participants in Türkiye and Cyprus (2025).
- TADD (a Turkish NGO implementing initiatives across Africa) scholarship & stipend for their Africa-oriented leadership & project development education (2026).

PUBLICATIONS & CONFERENCE PRESENTATIONS

Chawatama, K., & Tunca, B. (2026). Cross-Species Single-Cell Analysis Reveals Potential Drug Resistance Mechanisms in MASH vs Viral HCC. *19th International Symposium on Health Informatics and Bioinformatics (HIBIT 2026).* **Oral Presentation**

Chawatama, K., Tekin, C., Ercelik, M., & Tunca, B. (2026). Cross-Species Single-Cell Analysis Reveals Hepatocyte-Driven Reprogramming of the Tumor Immune Microenvironment in NASH vs Viral HCC. *European Journal of Immunology, 56*(S1), P07, 40, e70216. https://doi.org/10.1002/eji.70216 *(yEFIS 2026)*. **Poster**

Chawatama, K., & Tunca, B. (2026). Characterizing Tumor Microenvironment Transcriptome Variations in NASH and Viral Hepatocellular Carcinomas for Targeted Therapy. *18th International Symposium on Health Informatics and Bioinformatics (HIBIT 2025).* **Poster**

Amirak, E., Guler, M., **Chawatama, K.**, & Demirbag, M. (2023). Comprehensive Analysis of Promoter Methylation Effects on the GPRC5A Methylation Profile Across Various Cancer Types. *16th International Symposium on Health Informatics and Bioinformatics (HIBIT 2023).* **Poster**

RESEARCH SKILLS & METHODS

Experimental

- **Molecular Genetics Techniques:** DNA Isolation from Blood, Restriction Digestion, PCR, Agarose Gel Electrophoresis, Bacterial Culture & Cloning

Computational

- **Programming Languages:** R, Python, Nextflow, Bash, Arduino
- **NGS Data Analysis:** Bulk & Single Cell RNA-seq, TCR-seq, CITE-seq, Microarray, WES/WGS Processing & Variant Analysis, ChIP-seq, Methyl-seq
- **Frameworks & Environments:** Slurm (Linux HPC), Conda, Docker/Singularity, Nextflow, Snakemake, R-Shiny, Git, Arduino, Raspberry Pi
- **Machine Learning:** Basic Scikit-learn and TensorFlow

REFERENCES

Prof. Berrin Tunca
Uludağ University
btunca@uludag.edu.tr

Prof. Vicente Pelechano Garcia
Karolinska Institutet
vicent.pelechano@scilifelab.se

Dr. Matthias Zepper
(Sweden) National Genomics Infrastructure
matthias.zepper@scilifelab.se